



## Lesson 1.2 — Arithmetic Sequences

*Big idea: An arithmetic sequence adds the same number (the common difference  $d$ ) every step. That single fact unlocks every formula.*

### Quick review

**Common difference  $d$**  = any term minus the one before it. If  $d$  is constant, the sequence is arithmetic.

**Recursive:**  $a_1$  given,  $a_n = a_{n-1} + d$ .

**Explicit:**  $a_n = a_1 + (n - 1)d$ .

**Graph check:** Plotting  $(n, a_n)$  gives a straight line with slope  $d$ .

### Worked example

**Worked example.** A sequence starts 4, 11, 18, 25, ... Find the 30th term.

$$d = 11 - 4 = 7, a_1 = 4. a_{30} = 4 + (30-1)(7) = 4 + 203 = 207.$$

### Warm-ups

1. For the sequence 5, 12, 19, 26, ..., state  $d$  and write the explicit rule.
2. For the sequence 20, 17, 14, 11, ..., state  $d$  and find  $a_{10}$ .
3. Is 2, 4, 8, 16, ... arithmetic? Explain in one sentence.

### Practice

1. An arithmetic sequence has  $a_1 = -8$  and  $d = 3$ . Find  $a_{25}$ .
2. An arithmetic sequence has  $a_5 = 17$  and  $a_{12} = 38$ . Find  $d$  and  $a_1$ .
3. Write both a recursive and explicit rule for the sequence of even positive integers 2, 4, 6, ...

### Challenge

1. A movie theater has 15 seats in the front row and each row behind has 3 more seats than the one in front of it. How many seats are in the 20th row? How many total seats in the first 20 rows? (Hint: total =  $n(a_1 + a_n)/2$ .)
2. Sanjay opens a savings account with \$200 and adds \$35 at the end of each month. Write an explicit formula for the balance at the end of month  $n$  (ignore interest). After how many months does he first have more than \$1,000?



---

## Answer key — Lesson 1.2

*Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.*

---

### Warm-ups

1.  $d = 7$ ;  $a_n = 5 + (n-1)(7) = 7n - 2$ .
2.  $d = -3$ ;  $a_{10} = 20 + 9(-3) = -7$ .
3. No — the differences are 2, 4, 8, not constant. (It's geometric.)

### Practice

1.  $a_{25} = -8 + 24(3) = 64$ .
2.  $d = (38-17)/(12-5) = 3$ ;  $a_1 = 17 - 4(3) = 5$ .
3. Recursive:  $a_1 = 2$ ,  $a_n = a_{n-1} + 2$ . Explicit:  $a_n = 2n$ .

### Challenge

1. Row 20 has  $15 + 19(3) = 72$  seats. Total =  $20(15 + 72)/2 = 870$  seats.
2.  $a_n = 200 + 35n$  (balance at end of month  $n$ ).  $200 + 35n > 1000 \Rightarrow n > 22.86$ , so first exceeds \$1,000 at month 23 (balance \$985 at month 22, \$1020 at month 23).